

**Third Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ENGINEERING MATHEMATICS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

Use of Non – programmable calculator is permitted.

1. Solve any Two :

- (a) Determine whether $f(z) = e^z$ is analytic or not. If analytic find its derivative. 6(CO2)
- (b) Show that the function $u(x, y) = 3x^2y + x^2 - y^3 - y^2$ is harmonic function. Find a function $v(x, y)$ such that $u + iv$ is an analytic function. 6(CO2)
- (c) Find the image of the circle $|z| = a$ by the transformation $w = z + 3 + 4i$. 6(CO2)

2. Solve any Two :

- (a) Evaluate $\int_C \frac{dz}{(z^2 + 4)^3}$, where C is the circle $|z - i| = 2$. 6(CO2)
- (b) Evaluate by contour integration : $\int_0^{2\pi} \frac{d\theta}{17 - 8 \cos \theta}$. 6(CO2)
- (c) Find the expansion of $f(z) = 1 / (z - z^3)$ in the region $1 < |z - 1| < 2$. 6(CO2)

3. Solve any Two :

- (a) Solve the following Lagrange's linear equation :
 $p - q = \log(x + y)$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. 6(CO1)

(b) Solve : $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \cos x.$ 6(CO1)

(c) Using method of separation of variables, solve : $py^3 + qx^2 = 0$ where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}.$ 6(CO1)

4. Solve :

(a) Find Laplace Transform of following :

(i) $(2e^t + e^{-2t})t,$

(ii) $u(t - a),$ where $u(t)$ is a unit step function. 6(CO3)

(b) Solve by Laplace Transform method : $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 8y = \cos 2t,$ given that at $t = 0, y = 2$ and $\frac{dy}{dt} = 1.$ 6(CO3)

5. Solve :

(a) Find $Z\{e^{-iat}\}$ and hence deduce the value of $Z\{\cos at\}$ and $Z\{\sin at\}.$ 6(CO3)

(b) Solve by Z-transform method : $y_{n+2} + 2y_{n+1} + y_n = n$ where $y_0 = 0, y_1 = 0.$ 6(CO3)

